

5 Practical Numerical Problems with Answers

1. Calculate the V_{DIFF} for the Dominant State:

- **Given:**
 - $CAN_H = 3.5V$
 - $CAN_L = 1.5V$
 - **Solution:**
 $V_{DIFF} = CAN_H - CAN_L$
 $V_{DIFF} = 3.5V - 1.5V = 2.0V$
 - **Answer:** $V_{DIFF} = 2.0V$
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2. Data Frame Size:

- **Given:**
 - Data Field = 4 bytes = $4 \times 8 = 32$ bits
 - CRC = 15 bits
 - ID = 11 bits
 - Other Fields = 20 bits
 - **Solution:**
Total Frame Size = Data Field + CRC + ID + Other Fields
Total Frame Size = $32 + 15 + 11 + 20 = 78$ bits
 - **Answer:** Total frame size = 78 bits.
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3. Bus Arbitration:

- **Given:**
 - Node A: ID = 0x005 (Binary: 0000 0000 0101)
 - Node B: ID = 0x00A (Binary: 0000 0000 1010)
 - At the 3rd bit:
 - Node A sends 0
 - Node B sends 1
 - **Solution:**
In CAN arbitration, the dominant bit (0) overrides the recessive bit (1). Since Node A sends 0, it wins arbitration and continues transmitting.
 - **Answer:** Node A wins.
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4. Message Priority:

- **Given IDs:**
 - 0x001 (Binary: 0000 0000 0001)
 - 0x010 (Binary: 0000 0001 0000)
 - 0x100 (Binary: 0001 0000 0000)
- **Solution:**
Lower numerical value means higher priority in CAN.
Priority Order: $0x001 > 0x010 > 0x100$
- **Answer:** Priority order is 0x001, 0x010, 0x100.

5. Error Detection:

- **Given:**
 - Data = 0xAB
 - Transmitted CRC = 0x1F3
 - Receiver Calculated CRC = 0x1F2
- **Solution:**

CRC mismatch indicates an error in the message. The frame will be rejected, and an error frame will be sent.
- **Answer:** The frame is **not accepted** due to CRC mismatch.

10 Practical Short Questions with Answers

1. Explain the purpose of the arbitration mechanism in CAN.

The arbitration mechanism in CAN ensures that multiple nodes can attempt to transmit simultaneously, but only the one with the highest priority message (lowest identifier value) continues transmission. This is achieved without collisions, ensuring efficient communication.

2. What is the role of the differential signaling in CAN?

Differential signaling in CAN enhances noise immunity by using two wires (CAN_H and CAN_L). The signal is represented by the voltage difference between the wires, allowing CAN to reject common-mode noise.

3. Describe the dominant and recessive states in a CAN bus.

- **Dominant State:** When the voltage difference between CAN_H and CAN_L is greater than a certain threshold (e.g., 2V), representing a logical '0'.
 - **Recessive State:** When the voltage difference is minimal (near 0V), representing a logical '1'.
- The dominant state always overrides the recessive state in bus arbitration.

4. How does the CAN protocol handle errors?

CAN uses mechanisms like bit stuffing, acknowledgment, CRC checks, and error frames to detect and recover from errors. If an error is detected, the transmitting node re-transmits the message, ensuring reliability.

5. What are termination resistors, and why are they used in CAN networks?

Termination resistors (typically 120 ohms) are placed at both ends of the CAN bus to prevent signal reflections and ensure proper impedance matching, enabling reliable communication over long distances.

6. Why is CAN suitable for real-time applications?

CAN's priority-based arbitration and deterministic behavior ensure that high-priority messages are transmitted without delays, making it ideal for time-critical applications like automotive and industrial control systems.

7. Define the purpose of the CRC field in a CAN frame.

The CRC (Cyclic Redundancy Check) field ensures data integrity by detecting errors in the transmitted frame. The receiver calculates its CRC value and compares it with the transmitted CRC to verify correctness.

8. How does the "Wired OR" logic apply to CAN?

The CAN bus operates using "Wired OR" logic, where the dominant state ('0') overrides the recessive state ('1'). This allows arbitration to determine which node has the highest priority message without collisions.

9. What is the significance of the DLC field in a CAN frame?

The DLC (Data Length Code) field specifies the number of data bytes (0-8) in the CAN frame. It informs the receiver about the length of the payload, ensuring proper data handling.

10. What happens if two nodes transmit simultaneously in a CAN network?

Both nodes start transmitting simultaneously, but arbitration resolves the conflict. The node transmitting the message with the lower-priority identifier (higher numerical value) stops, allowing the higher-priority message to be transmitted.